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ENGINEERING



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CO 2 – ASSIGNMENT 1

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Course Code & CO Reference: CSA0920 – HashMap vs Hashtable

Assignment Set: CO 2 – Assignment 1

Question: HashMap vs Hashtable – Null Key/Value Handling

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GitHub Repository Link: <https://github.com/192424318-pradeep-1980/CSA0920-Java>

Question:

HashMap vs Hashtable

Write a Java program that creates both a HashMap and a Hashtable, and attempts to insert a null key or a null value into each. Display the outcome of each attempt (success or exception/error) to demonstrate that HashMap permits null keys/values while Hashtable does not.

Concepts: HashMap, Hashtable, null handling.

PROBLEM UNDERSTANDING:

The program creates a HashMap and a Hashtable and tries to insert null keys and null values into both. The purpose is to show that HashMap allows one null key and multiple null values, whereas Hashtable does not allow null keys or null values and throws an exception.

APPROACH / DESIGN NOTES:

- Create one HashMap and one Hashtable.
- Try to insert a null key and a null value into both.
- Use try-catch to handle the exception from Hashtable.
- Display whether each insertion is successful or causes an exception.

Alternative considered:

The same operations can be performed without try-catch, but the program would stop when Hashtable throws an exception. Therefore, try-catch is used to display all outcomes clearly.

SOURCE CODE:

```
import java.util.HashMap;

import java.util.Hashtable;

class Main {

    public static void main(String[] args)
```

```
{  
    HashMap<Integer, String> map = new HashMap<>();  
    Hashtable<Integer, String> table = new Hashtable<>();  
  
    try  
  
    {  
        map.put(null, "Hello");  
  
        System.out.println("HashMap null key: Success");  
    }  
catch (Exception e)  
  
    {  
        System.out.println("HashMap null key: Exception");  
    }  
  
    try  
  
    {  
        map.put(1, null);  
  
        System.out.println("HashMap null value: Success");  
    }  
catch (Exception e)
```

```

{
    System.out.println("HashMap null value: Exception");
}

try
{
    table.put(null, "Hello");

    System.out.println("Hashtable null key: Success");
}

catch (Exception e)
{
    System.out.println("Hashtable    null    key:    "    +
e.getClass().getSimpleName());
}

try
{
    table.put(1, null);

    System.out.println("Hashtable null value: Success");
}

catch (Exception e)

```

```
{  
    System.out.println("Hashtable null value: " +  
e.getClass().getSimpleName());  
}  
}  
}
```

Expected Output:

```
HashMap null key: Success  
HashMap null value: Success  
Hashtable null key: NullPointerException  
Hashtable null value: NullPointerException
```

Simple Explanation:

HashMap :

- It allows null key and null values

Hashtable :

- Does not allow null key or null values
- Hashtable gives NullPointerException when null is inserted.

TEST CASES:

S.No	Input/ Operation	Expected Output	Actual Output
1	HashMap put(null, "Hello")	Success	Success
2	HashMap put(1, null)	Success	Success
3	Hashtable put(null, "Hello")	NullPointerException	NullPointerException
4	Hashtable put(1, null)	NullPointerException	NullPointerException

SCREENSHOTS OF EXECUTION:

Test Case 1 - HashMap null key

```
HashMap null key: Success
```

Test Case 2 - HashMap null value

```
HashMap null value: Success
```

Test Case 3 - Hashtable null key

```
Hashtable null key: NullPointerException
```

Test Case 4 - Hashtable null value

```
Hashtable null value: NullPointerException
```

Execution output showing HashMap accepts null key/value and Hashtable throws NullPointerException.

REFLECTION:

I learned the difference between HashMap and Hashtable in Java. I understood that HashMap allows null keys and values, while Hashtable does not allow null keys or values.

DECLARATION:

I declare that the work presented in this assignment is original and completed by me. Any external resources or AI assistance used during the preparation of this assignment have been properly acknowledged.